

COUNTERFACTUAL REASONING

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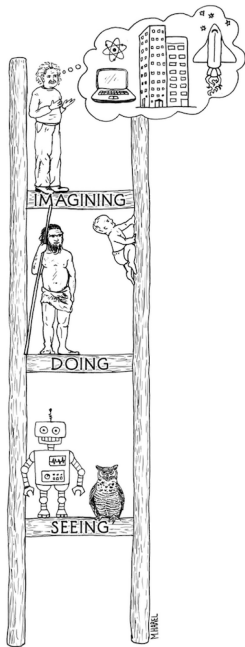
JUNE, 2026



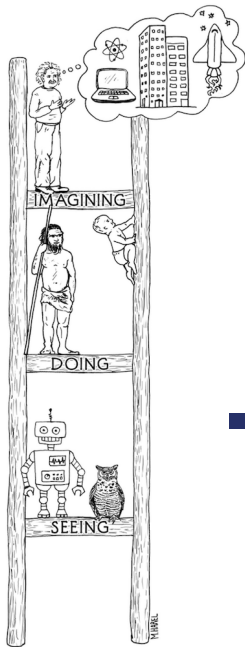
- 1 The Causal Hierarchy
- 2 Examples of Counterfactuals
- 3 Twin Networks
- 4 Single World Intervention Graphs (SWIGs)
- 5 Ancestral Multi World Network (AMWN)
- 6 Conclusion

1

THE CAUSAL HIERARCHY

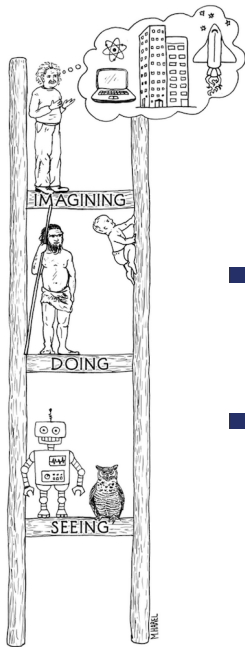


Pearl 2009



■ Associations

- ▶ What if I see...?
- ▶ Observational data
- ▶ Tools: statistics, correlation

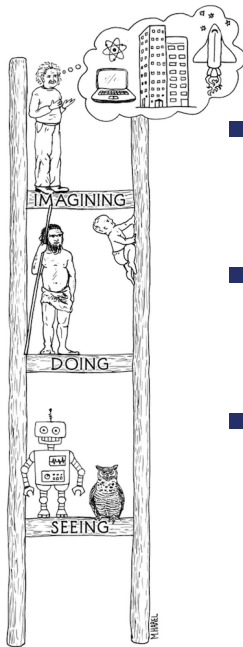


■ Interventions

- ▶ Questions: What if I do...? How?
- ▶ Interventional data
- ▶ Tools: RCTs, DAGs, do-calculus

■ Associations

- ▶ What if I see...?
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■ Counterfactuals

- ▶ Questions: What if I had done...? Why?
- ▶ Generating process (SCM)
- ▶ Tools: Twin Networks, SWIGs, AMWN

■ Interventions

- ▶ Questions: What if I do...? How?
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- ▶ What if I see...?
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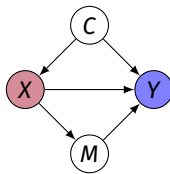
$\xi_C, \xi_X, \xi_M, \xi_Y$

$C := \xi_C$

$X := f_X(C, \xi_X)$

$M := f_M(X, \xi_M)$

$Y := f_Y(C, X, M, \xi_Y)$



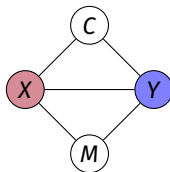
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■ Associations:

$\Pr(Y = y \mid X = x)$

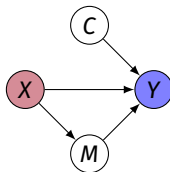
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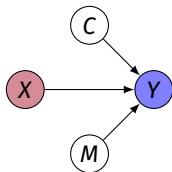
$\xi_C, \xi_X, \xi_M, \xi_Y$

$C := \xi_C$

$X := x$

$M := f_M(x', \xi_M)$

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■ Counterfactuals:

$\Pr(Y = y \mid \text{do}(X = x), \text{do}(M = m_{\text{do}(X=x')}))$

■ Interventions:

$\Pr(Y = y \mid \text{do}(X = x))$

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$\Pr(Y = y \mid X = x)$

$$\begin{aligned}
&\xi_C, \xi_X, \xi_M, \xi_Y \\
C &:= \xi_C \\
X &:= f_X(C, \xi_X) \\
M &:= f_M(X, \xi_M) \\
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\end{aligned}$$

$$\begin{aligned}
&\xi_C, \xi_X, \xi_M, \xi_Y \\
C &:= \xi_C \\
X_1 &:= x' \\
M &:= f_M(X_1, \xi_M) \\
X_2 &:= X \\
Y &:= f_Y(C, X_2, M, \xi_Y)
\end{aligned}$$

Counterfactuals:

$$\Pr(Y = y \mid \text{do}(X = x), \text{do}(M = m_{\text{do}(X=x')}))$$

2

EXAMPLES OF COUNTERFACTUALS

- Effect of the treatment on the treated:

$$\mathbb{E}(Y_{\text{do}(X=1)} \mid X = 1) - \mathbb{E}(Y_{\text{do}(X=0)} \mid X = 1)$$

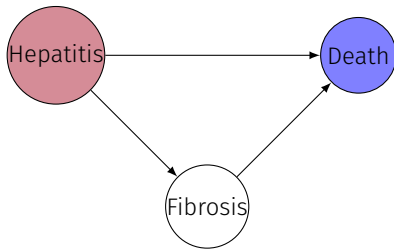
EXAMPLES OF COUNTERFACTUALS

- Effect of the treatment on the treated:
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- Natural Direct Effect

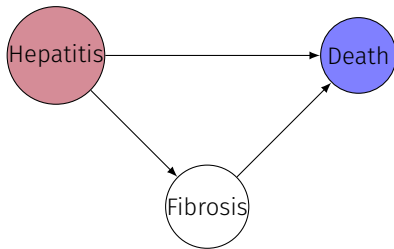
THE NATURAL DIRECT EFFECT



Direct Effect: How will a change in the exposure (Hepatitis) modify the outcome (Death) if the mediator (Fibrosis) is kept constant?

Robins and Greenland 1992; Pearl 2001

THE NATURAL DIRECT EFFECT



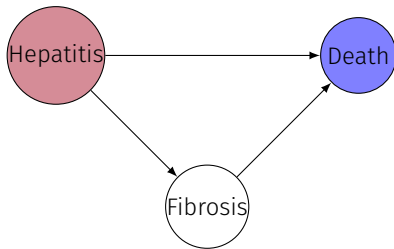
Direct Effect: How will a change in the exposure (Hepatitis) modify the outcome (Death) if the mediator (Fibrosis) is kept constant?

$$\begin{aligned}
 CDE(D, H_{0 \rightarrow 1}, F = f) &= \mathbb{E}(D \mid \text{do}(H = 1), \text{do}(F = f)) \\
 &\quad - \mathbb{E}(D \mid \text{do}(H = 0), \text{do}(F = f))
 \end{aligned}$$

This is called the controlled direct effect.

Robins and Greenland 1992; Pearl 2001

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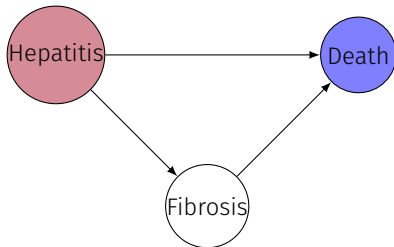
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 \end{aligned}$$

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Problem: This definition depends on the value of f . Thus there exists as many values of CDE as F has possible values.

Robins and Greenland 1992; Pearl 2001

THE NATURAL DIRECT EFFECT

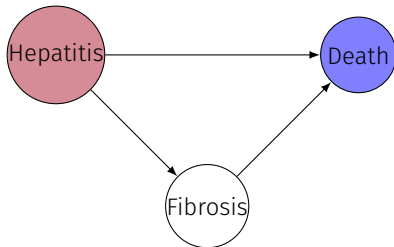


Direct Effect: How will a change in the exposure (Hepatitis) modify the outcome (Death) if the mediator (Fibrosis) is kept constant?

What is the natural value for the mediator Fibrosis?

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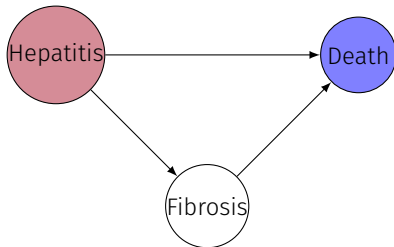


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What is the natural value for the mediator Fibrosis?

The natural value for Fibrosis is the value it would have taken had we not modified the exposure (Hepatitis)

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What is the natural value for the mediator Fibrosis?

The natural value for Fibrosis is the value it would have taken had we not modified the exposure (Hepatitis)

$$\begin{aligned}
 NDE(D, H_{0 \rightarrow 1}, F) &= \mathbb{E} (D \mid \text{do}(H = 1), \text{do}(F = f_{\text{do}(H=0)})) \\
 &\quad - \mathbb{E} (D \mid \text{do}(H = 0), \text{do}(F = f_{\text{do}(H=0)}))
 \end{aligned}$$

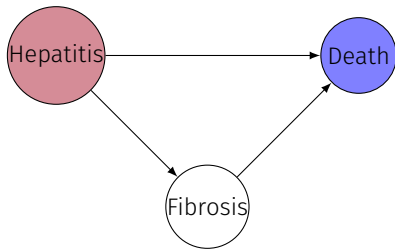
Robins and Greenland 1992; Pearl 2001

$$\xi_H, \xi_F, \xi_D \leftarrow \mathcal{U}(0, 1)$$

$$H := \xi_H$$

$$F := H \vee \xi_F$$

$$D := (H \vee F) \wedge \xi_D$$



TOY EXAMPLE

Observationnal Data:					
Noise			Variables		
ξ_H	ξ_F	ξ_D	H	F	D
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	1	0
1	0	1	1	1	1
1	1	0	1	1	0
1	1	1	1	1	1

Conditionnal average treatment effect (CATE):

$$\begin{aligned}
 & \mathbb{E}(D \mid H = 1, F = 0) - \mathbb{E}(D \mid H = 0, F = 0) \\
 &= 0 - 0 = 0
 \end{aligned}$$

TOY EXAMPLE

Noise			Interventional Data:					
			do ($H = 0, F = 0$)			do ($H = 1, F = 0$)		
ξ_H	ξ_F	ξ_D	H	F	D	H	F	D
0	0	0	0	0	0	1	0	0
0	0	1	0	0	0	1	0	1
0	1	0	0	0	0	1	0	0
0	1	1	0	0	0	1	0	1
1	0	0	0	0	0	1	0	0
1	0	1	0	0	0	1	0	1
1	1	0	0	0	0	1	0	0
1	1	1	0	0	0	1	0	1

Controlled direct effect (CDE)

$$\begin{aligned}
 & \mathbb{E}(D \mid \text{do}(H = 1, F = 0)) - \mathbb{E}(D \mid \text{do}(H = 0, F = 0)) \\
 &= \frac{1}{2} - 0 = \frac{1}{2}
 \end{aligned}$$

Counterfactual Data:

Noise			$\text{do}(H = 0, F = f_{\text{do}(H=0)})$			$\text{do}(H = 1, F = f_{\text{do}(H=0)})$		
ξ_H	ξ_F	ξ_D	H	F	D	H	F	D
0	0	0	0	0	0	1	0	0
0	0	1	0	0	0	1	0	1
0	1	0	0	1	0	1	1	0
0	1	1	0	1	1	1	1	1
1	0	0	0	0	0	1	0	0
1	0	1	0	0	0	1	0	1
1	1	0	0	1	0	1	1	0
1	1	1	0	1	1	1	1	1

Natural direct effect:

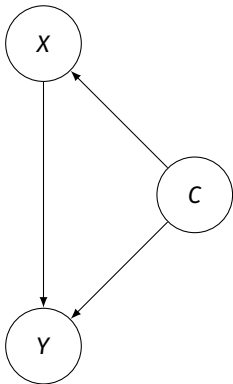
$$\begin{aligned} & \mathbb{E}(D \mid \text{do}(H = 1, F = f_{\text{do}(H=0)})) - \mathbb{E}(D \mid \text{do}(H = 0, F = f_{\text{do}(H=0)})) \\ &= \frac{1}{2} - \frac{1}{4} = \frac{1}{4} \end{aligned}$$

3

TWIN NETWORKS

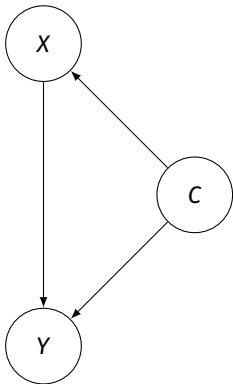
- One graph for the actual world

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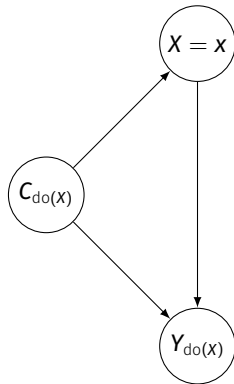
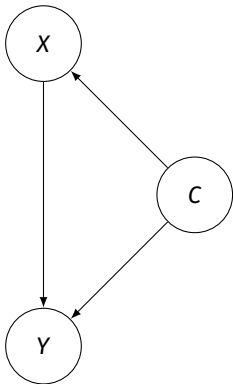
Balke and Pearl 1994

- One graph for the actual world
- One graph for the counterfactual world



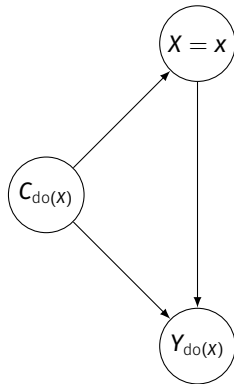
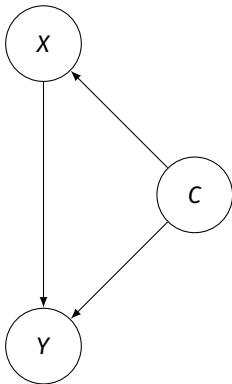
Balke and Pearl 1994

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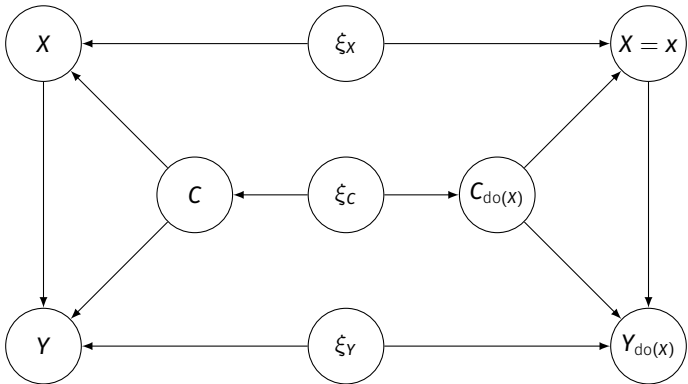
Balke and Pearl 1994

- One graph for the actual world
- One graph for the counterfactual world
- Common noise variables



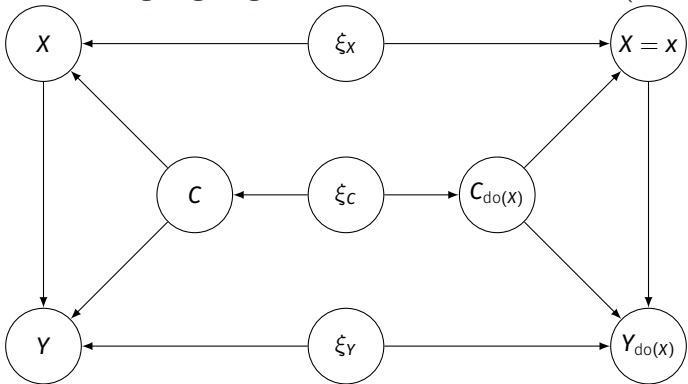
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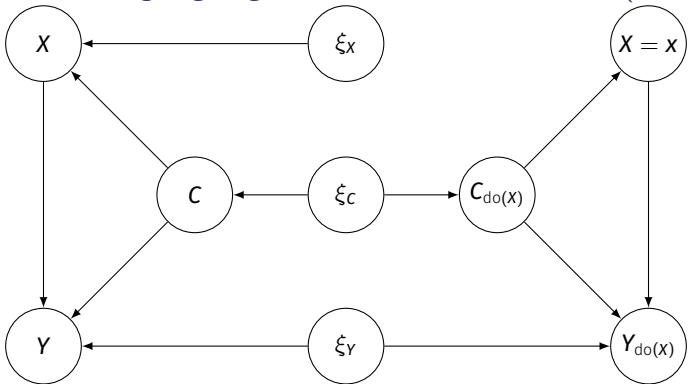
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- One graph for the actual world
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- Remove edges going in the intervened node ($\text{do}(X = x)$)



Balke and Pearl 1994

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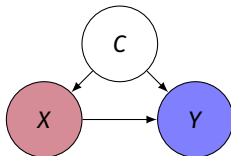
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What can Twin Networks be used for?

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SCM framework (Pearl)

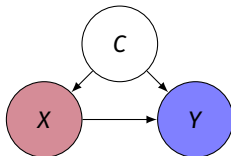
- Assume the graph
- Backdoor criterion: C blocks all backdoor paths from X to Y .



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Potential Outcome framework (Robins)

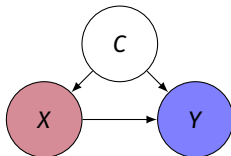
- Assume conditionnal exchangeability

$$Y_{\text{do}(X)} \perp\!\!\!\perp_{\text{Pr}} X \mid C$$

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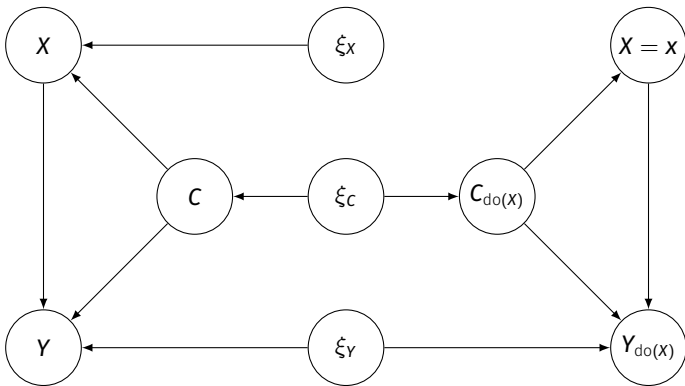
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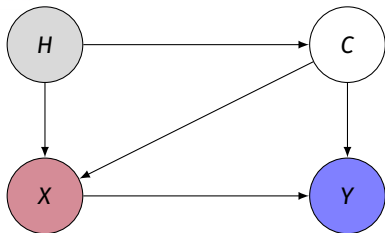
Average Treatment Effect:

$$\sum_c (\mathbb{E}(Y \mid X = 1, C = c) - \mathbb{E}(Y \mid X = 0, C = c)) \Pr(C = c)$$

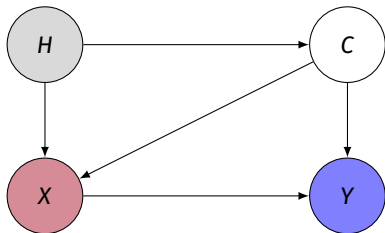
Conditional Exchangeability: $Y_{\text{do}(X)} \perp\!\!\!\perp_d X \mid C?$



TWIN NETWORKS: ANOTHER EXAMPLE

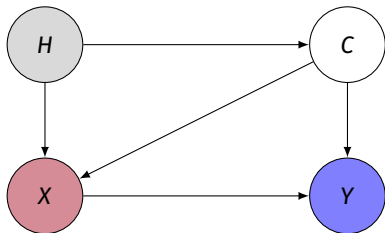


TWIN NETWORKS: ANOTHER EXAMPLE

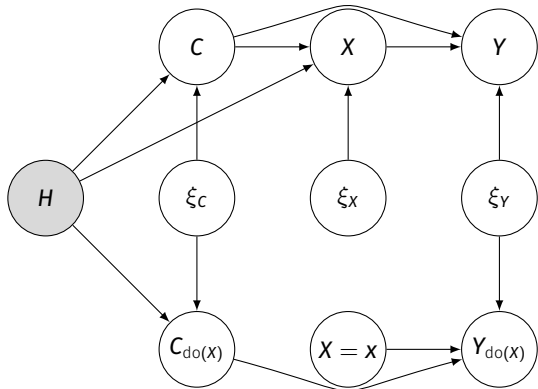


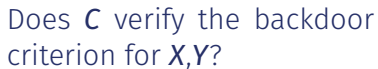
Does C verify the backdoor criterion for X, Y ?

TWIN NETWORKS: ANOTHER EXAMPLE



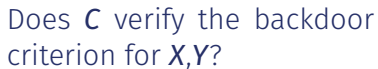
Does C verify the backdoor criterion for X, Y ?





$$Y_{\text{do}(X)} \perp\!\!\!\perp_d X \mid C?$$





$$Y_{\text{do}(X)} \perp\!\!\!\perp_d X \mid \mathcal{C}, \mathcal{C}_{\text{do}(X)}?$$



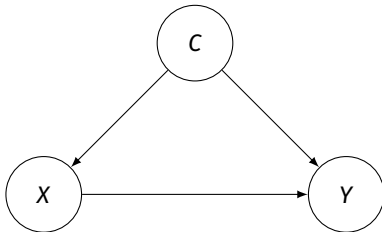
4

**SINGLE WORLD
GRAPHS (SWIGs)**

INTERVENTION

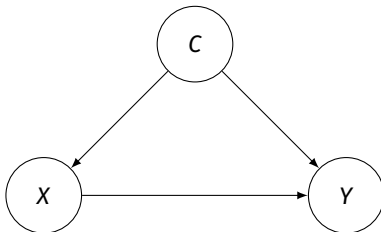
- One graph for both worlds

■ One graph for both worlds



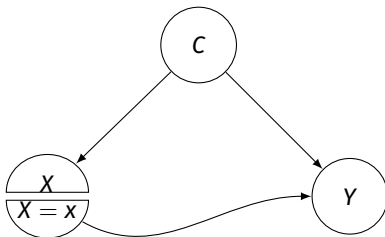
Richardson and Robins 2013

- One graph for both worlds
- Divide the intervened nodes ($\text{do}(X = x)$)

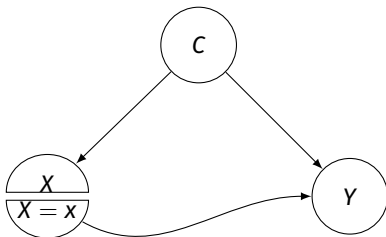


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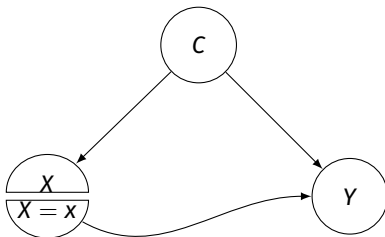
- One graph for both worlds
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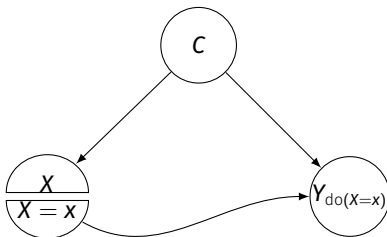
- One graph for both worlds
- Divide the intervened nodes ($\text{do}(X = x)$)
- Notice the edges

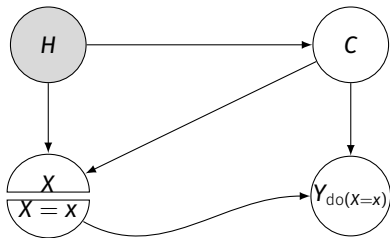
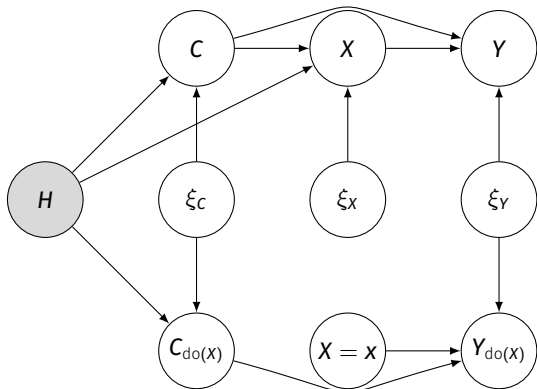


- One graph for both worlds
- Divide the intervened nodes ($\text{do}(X = x)$)
- Notice the edges
- Relabel nodes descendant of interventions



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$$Y_{do(X)} \perp\!\!\!\perp_d X \mid C?$$

The do-calculus is sound
(and complete for interventionnal queries)

Rule 1 $\Pr(y \mid \text{do}(x), z, w) = \Pr(y \mid \text{do}(x), w)$

if $(Y \perp\!\!\!\perp_d Z \mid X, W)_{\mathcal{G}_{\overline{X}}}$

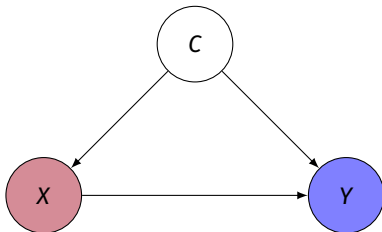
Rule 2 $\Pr(y \mid \text{do}(x, z), w) = \Pr(y \mid \text{do}(x), z, w)$

if $(Y \perp\!\!\!\perp_d Z \mid X, W)_{\mathcal{G}_{\overline{XZ}}}$

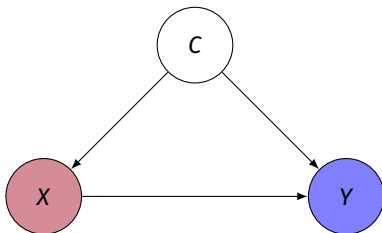
Rule 3 $\Pr(y \mid \text{do}(x, z), w) = \Pr(y \mid \text{do}(x), w)$

if $(Y \perp\!\!\!\perp_d Z \mid X, W)_{\mathcal{G}_{\overline{XZ(W)}}}$

THE BACKDOOR: AN APPLICATION OF THE DO-CALCULUS

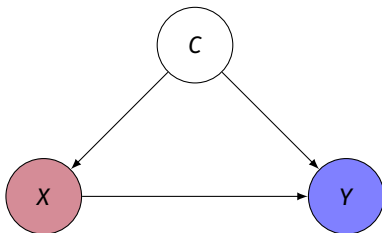


$$\text{ATE: } \mathbb{E}(Y \mid \text{do}(X = x_1)) - \mathbb{E}(Y \mid \text{do}(X = x_0))$$



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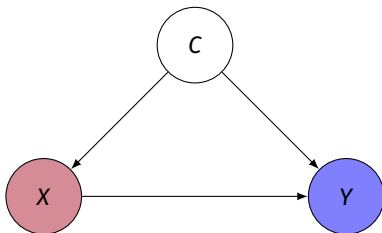
$$\Pr(Y \mid \text{do}(X))$$



$$\text{ATE: } \mathbb{E}(Y \mid \text{do}(X = x_1)) - \mathbb{E}(Y \mid \text{do}(X = x_0))$$

$$\Pr(Y \mid \text{do}(X)) = \sum_c \Pr(Y \mid \text{do}(X), C = c) \Pr(C = c \mid \text{do}(X))$$

THE BACKDOOR: AN APPLICATION OF THE DO-CALCULUS

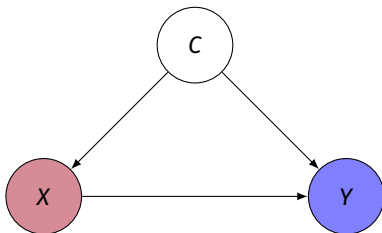


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$$\text{Rule 1 } \Pr(Y_{\text{do}(x)} \mid Z_{\text{do}(x)}, W_{\text{do}(x)}) = \Pr(Y_{\text{do}(x)} \mid W_{\text{do}(x)})$$

$$\text{if } (Y_{\text{do}(x)} \perp\!\!\!\perp_d Z_{\text{do}(x)} \mid W_{\text{do}(x)})_{\mathcal{G}_{\text{do}(x)}}$$

$$\text{Rule 2 } \Pr(Y_{\text{do}(x,z)} \mid W_{\text{do}(x,z)}) = \Pr(Y_{\text{do}(x)} \mid Z_{\text{do}(x)}, W_{\text{do}(x)})$$

$$\text{if } (Y_{\text{do}(x,z)} \perp\!\!\!\perp_d Z_{\text{do}(x,z)} \mid W_{\text{do}(x,z)})_{\mathcal{G}_{\text{do}(x,z)}}$$

$$\text{Rule 3 } \Pr(Y_{\text{do}(x,z)} \mid W_{\text{do}(x,z)}) = \Pr(Y_{\text{do}(x)} \mid W_{\text{do}(x)})$$

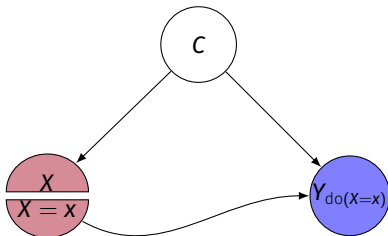
$$\text{if } (Y_{\text{do}(x,z_1)}, W_{\text{do}(x,z_1)} \perp\!\!\!\perp_d Z_1)_{\mathcal{G}_{\text{do}(x,z_1)}}$$

$$\text{and } (Y_{\text{do}(x,z_1)} \perp\!\!\!\perp_d Z_2 \mid W_{\text{do}(x,z_1)})_{\mathcal{G}_{\text{do}(x,z_1)}}$$

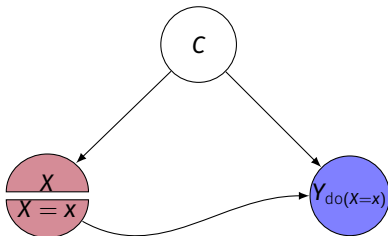
$$\text{where } Z_1 = Z \setminus \text{An}(W, \mathcal{G}_{\text{do}(x)}) \text{ and } Z_2 = Z \cap \text{An}(W, \mathcal{G}_{\text{do}(x)})$$

Malinsky, Shpitser, and Richardson 2019

THE BACKDOOR: AN APPLICATION OF THE PO-CALCULUS

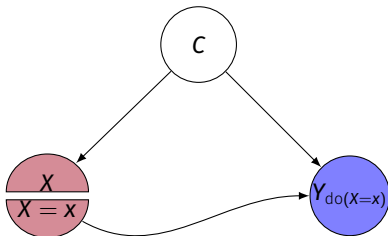


$$\text{ATE: } \mathbb{E} (Y_{\text{do}(X=x_1)}) - \mathbb{E} (Y_{\text{do}(X=x_0)})$$



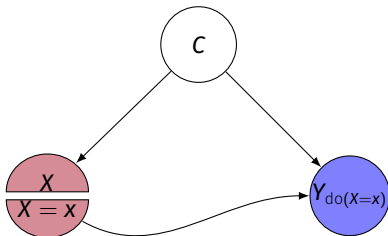
$$\text{ATE: } \mathbb{E} (Y_{\text{do}(X=x_1)}) - \mathbb{E} (Y_{\text{do}(X=x_0)})$$

$$\Pr (Y_{\text{do}(X)})$$



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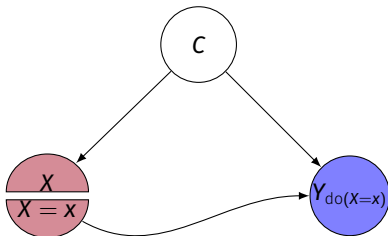
$$\Pr (Y_{\text{do}(X)}) = \sum_c \Pr (Y_{\text{do}(X)} \mid C_{\text{do}(X)} = c) \Pr (C_{\text{do}(X)} = c)$$



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SWIGs: Single World Interventional Graphs

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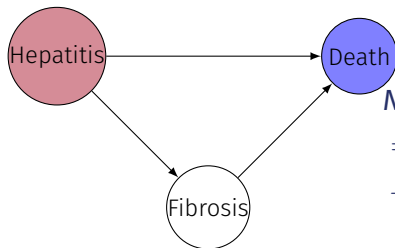
What is the effect of in the population that would die if they do not take it?

$$\mathbb{E} (Y_{\text{do}(X=1)} \mid Y_{\text{do}(X=0)} = 0)$$

SWIGs: Single World Interventional Graphs

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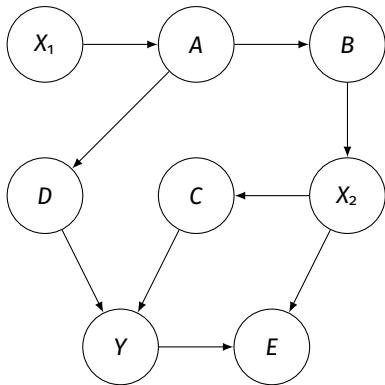


$$\begin{aligned}
 &NDE(D, H_{0 \rightarrow 1}, F) \\
 &= \mathbb{E}(D \mid \text{do}(H = 1, F = f_{\text{do}(H=0)})) \\
 &- \mathbb{E}(D \mid \text{do}(H = 0, F = f_{\text{do}(H=0)}))
 \end{aligned}$$

5

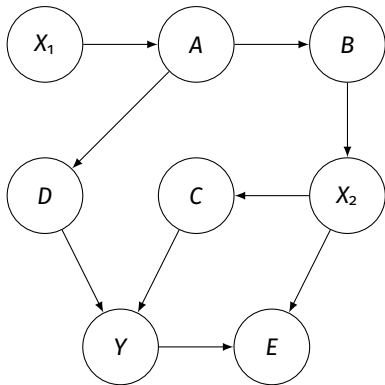
**ANCESTRAL MULTI WORLD NETWORK
(AMWN)**

ANCESTOR OF A COUNTERFACTUAL VARIABLE



$$\begin{aligned}
 An(Y_{do(\mathfrak{x})}) &= \{W_{do(\mathfrak{z})} \mid \\
 W &\in An(Y, G_{\overline{\mathfrak{X}}}) \setminus \mathfrak{X}, \\
 \mathfrak{z} &= \mathfrak{x} \cap An(W, G_{\overline{\mathfrak{X}}})\}
 \end{aligned}$$

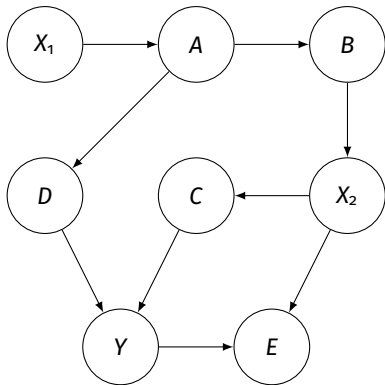
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$$An(Y_{do(X_1, X_2)})?$$

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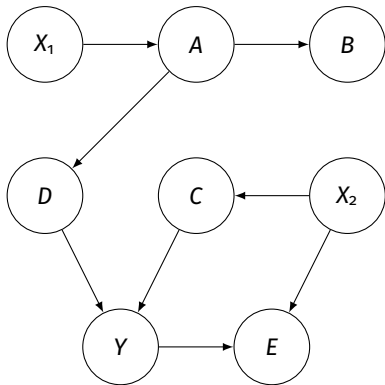


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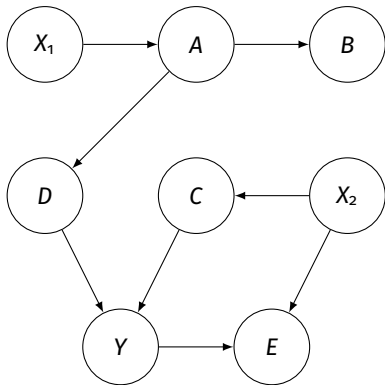


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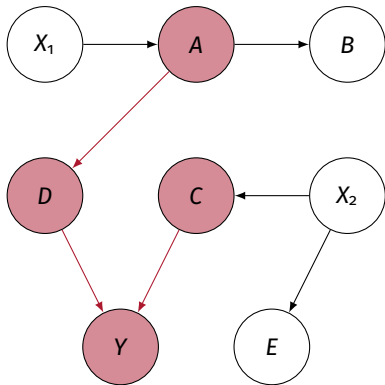
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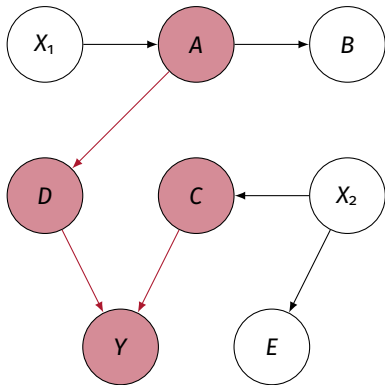
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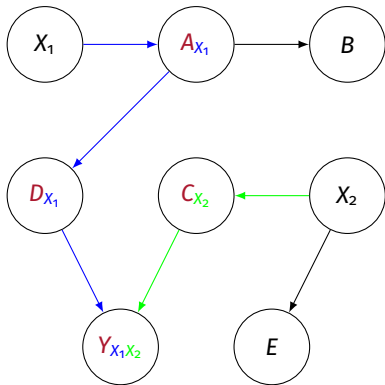
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$$An(Y_{do(X_1, X_2)})?$$

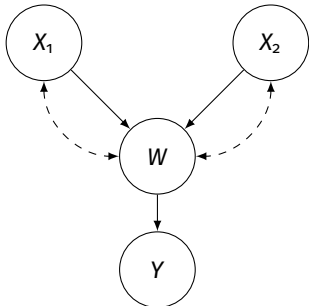
$$\cdot G_{\overline{X_1, X_2}} \cdot An(Y, G_{\overline{X_1, X_2}}) \setminus \{X_1, X_2\} \cdot \{x_1, x_2\} \cap An(W, G_{\overline{X_1, X_2}})$$

For a query on counterfactual variables:

- Add all ancestors of the counterfactual variables
- Add edges witnessing the ancestrality
- Add the hidden variables
- Add hidden confounders between same variables

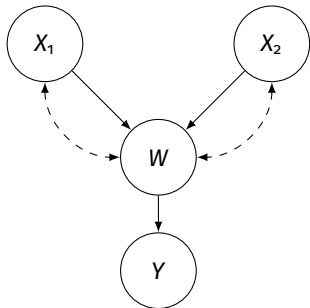
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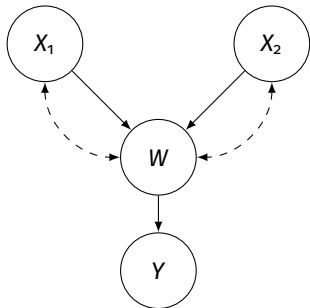
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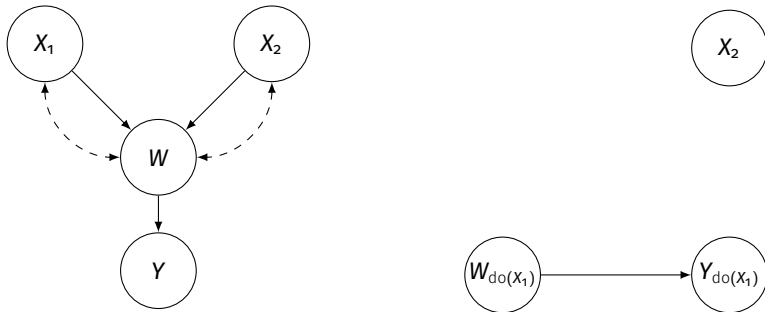


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Correa and Bareinboim 2025

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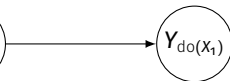
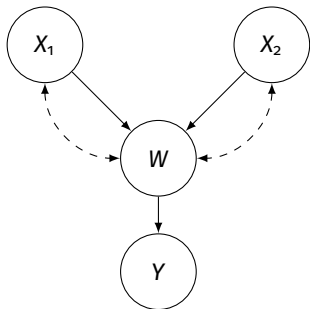
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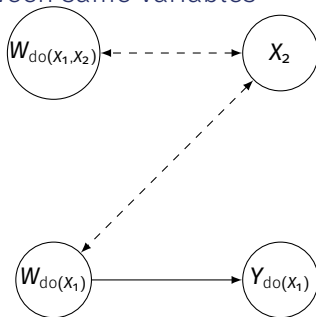
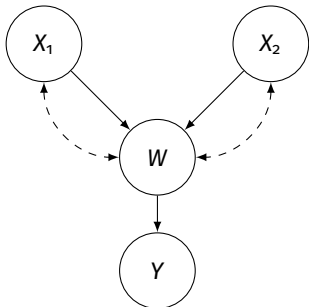
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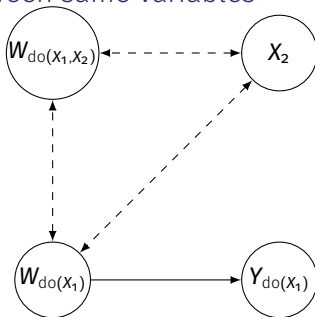
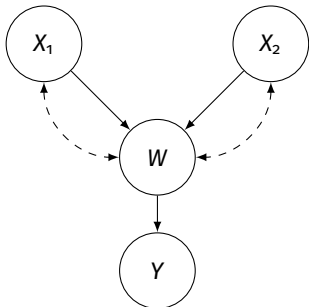
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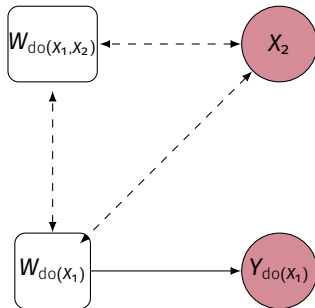
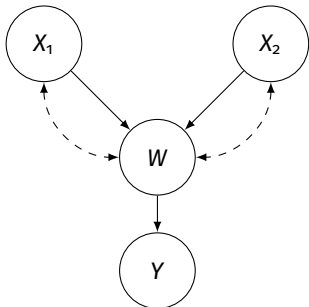
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The ctf-calculus is sound and complete for counterfactual queries.

$$\text{Rule 1 } \Pr(y_{T_*X}, x_{T_*}, w_*) = \Pr(Y_{T_*}, x_{T_*}, w_*)$$

$$\text{Rule 2 } \Pr(y_r \mid x_t, w_*) = \Pr(y_r \mid w_*)$$

$$\text{if } (Y_r \perp\!\!\!\perp_d X_t \mid W_*)_{\mathcal{G}_A}$$

$$\text{Rule 3 } \Pr(Y_{xz}, w_*) = \Pr(y_z, w_*)$$

$$\text{if } X \cap \text{An}(Y, \mathcal{G}_{\bar{Z}}) = \emptyset$$

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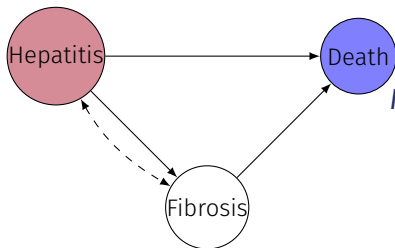
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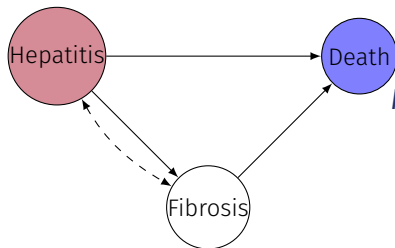
$$\text{if } X \cap \text{An}(Y, \mathcal{G}_{\bar{Z}}) = \emptyset$$

It transforms counterfactual queries into interventional queries. You might still need to use the do-calculus afterwards to transform the interventional queries into observational queries.

THE NATURAL DIRECT EFFECT



$$\begin{aligned}
 &NDE(D, H_{0 \rightarrow 1}, F) \\
 &= \mathbb{E}(D \mid \text{do}(H = 1, F = f_{\text{do}(H=0)})) \\
 &- \mathbb{E}(D \mid \text{do}(H = 0, F = f_{\text{do}(H=0)}))
 \end{aligned}$$



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 \end{aligned}$$

How do you identify

$$\Pr(D \mid \text{do}(H = 1, F = f_{\text{do}(H=0)})) = \Pr(d_{h'w_h})?$$

$$\Pr(d_{h'F_h})$$

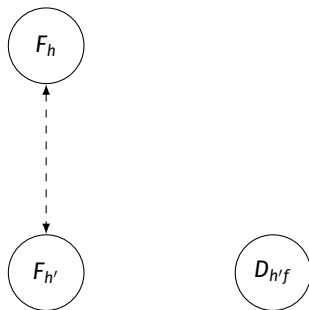
$$\Pr(d_{h'F_h}) = \sum_f \Pr(d_{h'F_h}, f)$$

$$\Pr(d_{h'F_h}) = \sum_f \Pr(d_{h'F_h}, f)$$

$$(R1) = \sum_f \Pr(d_{h'f}, f_h)$$

$$\begin{aligned}\Pr(d_{h'F_h}) &= \sum_f \Pr(d_{h'F_h}, f) \\ (R1) &= \sum_f \Pr(d_{h'f}, f_h) \\ &= \sum_f \Pr(d_{h'f} \mid f_h) \Pr(f_h)\end{aligned}$$

$$\begin{aligned}
 \Pr(d_{h'F_h}) &= \sum_f \Pr(d_{h'F_h}, f) \\
 (R1) &= \sum_f \Pr(d_{h'f}, f_h) \\
 &= \sum_f \Pr(d_{h'f} | f_h) \Pr(f_h) \\
 (R2) &= \sum_f \Pr(d_{h'f} | f_{h'}) \Pr(f_h)
 \end{aligned}$$



$$(D_{h'f} \perp\!\!\!\perp_d F_h, F_{h'})_{\mathcal{G}_A}$$

CTF-CALCULUS: EXAMPLE OF CALCULATIONS

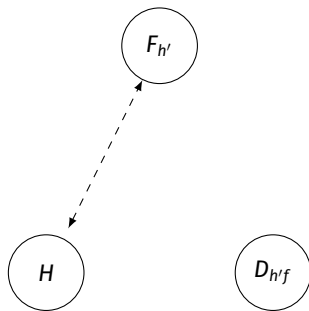
$$\Pr(d_{h'f_h}) = \sum_f \Pr(d_{h'f_h}, f)$$

$$(R1) = \sum_f \Pr(d_{h'f}, f_h)$$

$$= \sum_f \Pr(d_{h'f} | f_h) \Pr(f_h)$$

$$(R2) = \sum_f \Pr(d_{h'f} | f_{h'}) \Pr(f_h)$$

$$(R2) = \sum_f \Pr(d_{h'f} | f_{h'} h') \Pr(f_h)$$



$$(D_{h'f} \perp\!\!\!\perp_d H | F_{h'})_{\mathcal{G}_A}$$

$$\Pr(d_{h'F_h}) = \sum_f \Pr(d_{h'F_h}, f)$$

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$$= \sum_f \Pr(d_{h'f} | f_h) \Pr(f_h)$$

$$(R2) = \sum_f \Pr(d_{h'f} | f_{h'}) \Pr(f_h)$$

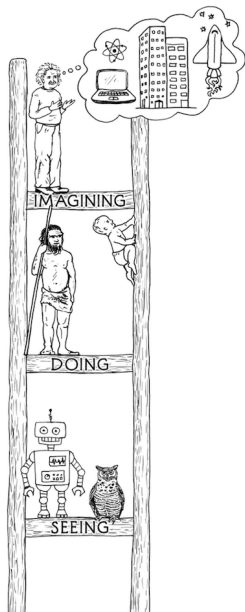
$$(R2) = \sum_f \Pr(d_{h'f} | f_{h'} h') \Pr(f_h)$$

$$(R1) = \sum_f \Pr(d_{h'f} | f, h') \Pr(f_h)$$

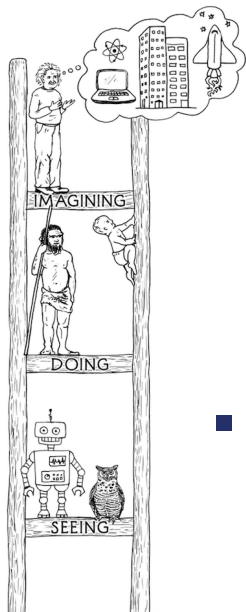
$$(R1) = \sum_f \Pr(d | f, h') \Pr(f_h)$$

6

CONCLUSION

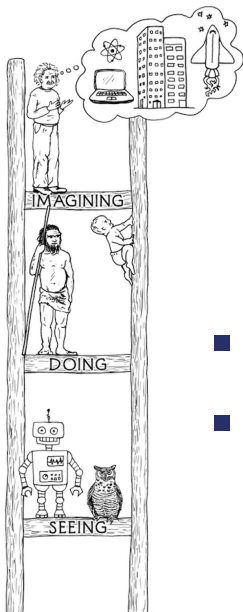


No assumptions



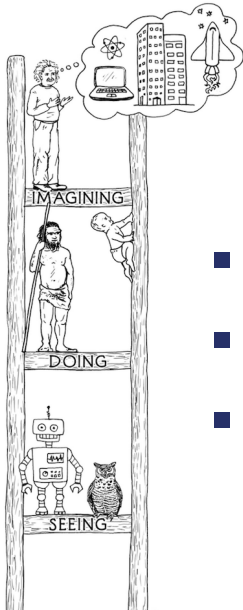
No assumptions

- Associations
 - Observationnal data



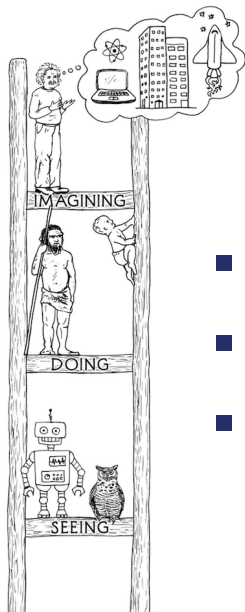
No assumptions

- Interventions
 - ▶ Interventionnal data
- Associations
 - ▶ Observationnal data



No assumptions

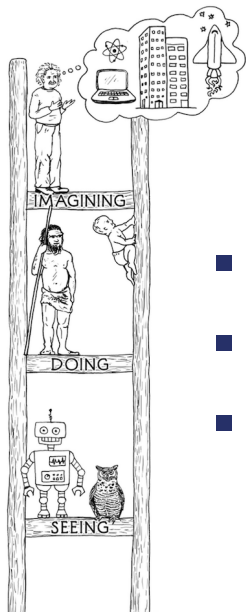
- Counterfactuals
 - ▶ Generating process
- Interventions
 - ▶ Interventional data
- Associations
 - ▶ Observational data



No assumptions

Assumptions: DAG

- **Counterfactuals**
 - ▶ Generating process
- **Interventions**
 - ▶ Interventionnal data
- **Associations**
 - ▶ Observationnal data



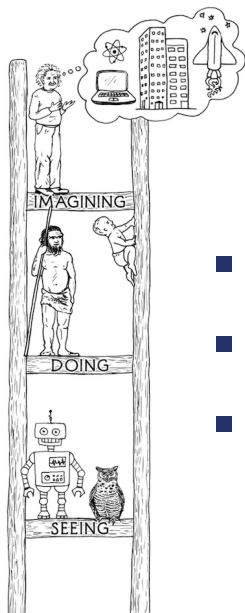
Counterfactual Reasoning

No assumptions

Assumptions: DAG

- **Counterfactuals**
 - ▶ Generating process
- **Interventions**
 - ▶ Interventional data
- **Associations**
 - ▶ Observational data

do-calculus



No assumptions

- **Counterfactuals**

- ▶ Generating process

- **Interventions**

- ▶ Interventional data

- **Associations**

- ▶ Observational data

Assumptions: DAG

ctf-calculus

do-calculus

THANK YOU FOR YOUR ATTENTION!

ANY QUESTIONS?





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